

Reading Note: *Game Theory with Simulation of Other Players*

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Abstract

This is an informal reading note on the IJCAI 2023 paper *Game Theory with Simulation of Other Players* by Kovařík, Oesterheld, and Conitzer. The paper studies games in which one player can pay a cost c to simulate the other player and learn their action before responding. I summarize the model and the main structural and welfare results, work through the trust-game example, and end with several questions that came to mind while reading. I wrote this note after attending Prof. Conitzer’s talk at the 37th Stony Brook Game Theory Festival in July 2026. All results discussed here are due to the authors, and any errors in interpretation are mine.

1 Motivation

Classical game theory does not require its players to be human, but AI agents make some previously unusual assumptions worth studying. One is that an agent might be simulatable. If enough of its implementation is available, another party may be able to run it in a sandbox and predict its action. The paper treats this prediction as accurate, while also charging the simulator a cost.

At first I expected this ability simply to make cooperation easier. In a trust game, after all, a player would no longer have to rely entirely on the other side’s incentives. The paper confirms this intuition for a broad class of trust games, but it also shows why it is not true in general. Simulation changes the information structure of a game, and more information is not always good for the interaction as a whole.

2 The Model

Begin with a finite two-player normal-form game

$$G = (A_1 \times A_2, u).$$

Player 1 is the *simulator* and Player 2 is the *simulated player*. For a fixed best-response policy ψ_{br} , Player 1 receives one additional action S . Choosing S means paying c , observing Player 2’s realized action b , and then best-responding to it:

$$u_1(S, b) = u_1(\text{br}, b) - c, \quad u_2(S, b) = \mathbb{E}_{a \sim \psi_{\text{br}}(b)} u_2(a, b).$$

The resulting game, denoted G_{sim}^c , is still a finite normal-form game, so ordinary Nash-equilibrium analysis applies after the new row S is added.

Two assumptions are especially important:

- Simulation reveals Player 2’s *realized action*, not merely the mixed strategy. Thus the simulator is assumed to predict the outcome of Player 2’s randomization. A later AAMAS 2025 paper studies what changes when this randomization is unpredictable.
- The simulated player either does not know that simulation occurred or cannot react to that fact. After observing b , the simulator is also required to best-respond, so the model rules out non-credible threats.

The first assumption is much stronger than it initially appears. If an LLM is sampled with nonzero temperature, querying an identical copy usually produces another sample, not advance knowledge of the original agent’s realized action. This distinction matters for the experiment proposed below.

3 Structural Results

Extreme costs (Prop. 7). If $c < 0$, simulation is subsidized and becomes a strongly dominant action for Player 1. Equilibrium play then effectively takes the form of a pure-commitment game with Player 2 as the Stackelberg leader. If c is sufficiently large, S is strictly dominated and G_{sim}^c has the same equilibria as G . The interesting behavior therefore occurs at intermediate costs.

Value of information (Lemma 10). Let $\text{VoIS}(\pi_2)$ denote the gain from best-responding to Player 2’s realized action rather than choosing a best response only to the mixed strategy π_2 :

$$\text{VoIS}(\pi_2) = \sum_{b \in A_2} \pi_2(b) \max_{a \in A_1} u_1(a, b) - \max_{a \in A_1} \sum_{b \in A_2} \pi_2(b) u_1(a, b).$$

If Player 1 mixes between simulation and a non-simulation action in equilibrium, indifference requires

$$\text{VoIS}(\pi_2) = c.$$

I found this a useful way to read the equilibrium: Player 2’s strategy must make the information gained from simulation exactly worth its price.

The trust-game calculation. For the paper’s running example, the simulation game can be written as

	C	D
T	$(25, 25)$	$(-100, 150)$
WO	$(0, 0)$	$(0, 0)$
S	$(25 - c, 25)$	$(-c, 0)$.

Suppose Alice mixes between trusting with probability x and simulating with probability $1 - x$. Bobble is indifferent between cooperation and defection when

$$25 = 150x,$$

so $x = 1/6$ and Alice simulates with probability $5/6$. Next let q be Bobble’s probability of defection. Alice is indifferent between trusting and simulating when

$$25 - 125q = 25 - 25q - c,$$

which gives

$$q = \frac{c}{100}.$$

For the paper’s choice $c = 7$, this yields

$$\pi_A(S) = \frac{5}{6}, \quad \pi_B(D) = \frac{7}{100}.$$

Piecewise constant/linear trajectories (Prop. 11). Between finitely many breakpoint values of c , the simulator’s equilibrium strategy remains constant while the simulated player’s strategy changes linearly with c . The trust-game calculation makes the claim concrete: Alice’s probability of simulation remains $5/6$, whereas Bobble’s probability of defection is $c/100$.

This was the most surprising result for me. Cheaper simulation need not produce more simulation. Here Alice’s mixing probability is fixed by the need to keep Bobble indifferent. A change in cost is instead absorbed by a change in Bobble’s behavior.

Computation (Thms. 2–3). For generic payoffs and sufficiently cheap simulation, all equilibria are pure or have supports of size two, and all equilibria can be found in $O(|A|)$ time. In this setting, adding simulation gives the game enough structure to make equilibrium computation easier. For a general normal-form game, however, it is co-NP-hard to determine whether simulation introduces an equilibrium that weakly Pareto-dominates every equilibrium of the original game.

4 Welfare: When Does Simulation Help?

Generalized trust games (Thm. 4). A generalized trust game is one in which every pure-commitment equilibrium, with Player 2 leading, strictly Pareto-dominates every Nash equilibrium of the original game. In other words, the players could both do better if Player 2 could credibly commit. The theorem shows that, for sufficiently small positive c , the simulation game has an equilibrium using S that strictly Pareto-improves upon every equilibrium of G . In this class of games, simulation provides the credible information needed to support cooperation.

Outside trust games, there is no single welfare conclusion:

- **Zero-sum games (Prop. 21).** Simulation weakly benefits the simulator and weakly harms the simulated player relative to the original equilibrium value. The inequalities need not be strict.
- **Commitment game (Ex. 22).** Free simulation creates an additional equilibrium that is worse for the simulator and makes the original simulator-favored equilibrium fail trembling-hand perfection. With subsidized simulation, the original equilibrium can disappear entirely. Merely having an additional option can therefore change which outcomes remain strategically stable.
- **Privacy-dependent cooperation (Ex. 23).** In the joint-project example, cheap simulation lets Player 1 learn Player 2’s password and steal the project’s proceeds. Anticipating this, Player 2 refuses to collaborate. This example shows that transparency can be harmful when cooperation depends on one party retaining private information.
- **An intermediate optimum (Ex. 25).** There are games in which a Pareto-optimal equilibrium outcome requires simulation to be neither free nor prohibitively expensive. This suggests that the simulation cost can itself act as a mechanism-design parameter.

My main takeaway is that simulation should not be treated as an inherently cooperative technology. It helps when the main obstacle is uncertainty about whether the other player will honor a cooperative action. It can hurt when the interaction depends on privacy, or when observing the other player

changes the commitment structure in an unfavorable way. The relevant question is therefore not simply whether an agent can be simulated, but what role information plays in the original game.

5 Questions and a Small Experiment

1. **LLM agents and equilibrium predictions.** I would like to implement the trust game with LLM agents and compare their observed behavior with the equilibrium probabilities above. I would compare three conditions: no simulation, perfect revelation of the realized action, and sample-based “simulation” obtained by querying a copy of the other agent. I am especially interested in the cooperation, defection, and simulation rates, and in their sensitivity to how the game is described.

This would not be a direct test of the Nash-equilibrium theorem: prompted LLM agents need not maximize expected utility or converge to equilibrium. I would treat it instead as a behavioral comparison between the theoretical benchmark and the decisions produced by current agents.

2. **What counts as simulation for an LLM?** The original model reveals the realized action. For an LLM with nonzero temperature, querying a copy normally reveals only another sample. This is closer to the unpredictable-randomization setting studied in the AAMAS 2025 follow-up. Choosing the right operational definition therefore seems like a modeling question rather than a minor implementation detail.
3. **Who benefits from being legible?** In the trust game, Bobble benefits from making its behavior simulatable. In the password game, the simulated player is harmed by the loss of privacy. For real AI systems, I would like to understand whether there is a useful way to distinguish interactions limited mainly by trust from those limited mainly by privacy.

References

- [1] V. Kovařík, C. Oesterheld, and V. Conitzer. Game Theory with Simulation of Other Players. In *Proceedings of IJCAI 2023*, pp. 2800–2807, 2023. <https://doi.org/10.24963/ijcai.2023/312>.
- [2] V. Kovařík, N. Sauerberg, L. Hammond, and V. Conitzer. Game Theory with Simulation in the Presence of Unpredictable Randomisation. In *Proceedings of AAMAS 2025*, pp. 1191–1199, 2025.